



YEAR 7 MATHEMATICS · CHAPTER 10

# The mean and mode

Lesson 5 of 8 ●●●●●●○

**In this lesson you will learn to:**

- calculate the mean and the mode of a set of data
- use the mean to find a missing value
- decide whether the mean or the mode better describes the data

Name \_\_\_\_\_

Class \_\_\_\_\_

Teacher \_\_\_\_\_

## Before you start

You will need

Pencil

Calculator

Syllabus: MA4-DAT-C-02: analyses simple datasets using measures of centre, range and shape of the data

### WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

### YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1 Work through Level 1, then Level 2, then Level 3. Try the extension if you finish early.

- 2 At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

## Lesson map

| Page | What is on it                                                                     | Done                     |
|------|-----------------------------------------------------------------------------------|--------------------------|
| 3    | Learning intentions, key terms, notes and Level 1 teacher examples                | <input type="checkbox"/> |
| 4    | Level 1: <b>YOU DO</b> practice questions                                         | <input type="checkbox"/> |
| 5    | Level 1 and 2: <b>WE DO</b> teacher examples and <b>YOU DO</b> practice questions | <input type="checkbox"/> |
| 6    | Level 2: <b>YOU DO</b> practice questions                                         | <input type="checkbox"/> |
| 7    | Level 3: <b>WE DO</b> teacher examples and <b>YOU DO</b> practice questions       | <input type="checkbox"/> |
| 8    | Level 3: <b>YOU DO</b> practice questions                                         | <input type="checkbox"/> |
| 9    | Extension questions and the tear-off exit ticket                                  | <input type="checkbox"/> |
| 10   | Lesson summary: a worked example for every level                                  | <input type="checkbox"/> |

### YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

1  $4 + 7 + 5 + 8 + 6$

3  $52 \div 7$  to 2 decimal places

2  $30 \div 5$

4 Most common value: 2, 5, 5, 8, 5, 1



### Acknowledgement of artwork

The artwork used throughout this booklet series was created by Marni Tuala. We thank Marni for sharing this artwork with our school community.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
10.05

# The mean and mode

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- calculate the mean and the mode of a set of data
- use the mean to find a missing value
- decide whether the mean or the mode better describes the data

## Success criteria

I can:

- ☐ find the **mean** by adding the values and dividing by how many there are
- ☐ find the **mode**, and tell when there is no mode or more than one
- ☐ work backwards from the mean to find a missing value

**Key terms** Write the definition as your teacher explains it.

**Mean**

**Mode**

Notes

## Level 1 · Getting started

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Find the **mean** of: 4, 7, 5, 8, 6

**Example 2** Find the **mode** of: 3, 9, 4, 7, 4, 8, 5

**YOU DO**

On your own: Set A. Find the mean of each set of numbers.

**1** 2, 4, 6**6** 6, 6, 9, 3**2** 5, 7, 9**7** 2, 8, 5, 7, 3**3** 1, 3, 5, 7**8** 12, 15, 18**4** 10, 20, 30**9** 4, 9, 2, 5, 10**5** 3, 5, 4, 8**10** 7, 11, 9, 13

**YOU DO**

On your own: Set B. Find the mode. Write “no mode” if there isn’t one.

1 2, 3, 3, 5, 8

6 4, 9, 7, 9, 2, 5, 9

2 1, 4, 4, 6, 9

7 1, 2, 3, 4, 5

3 7, 2, 5, 2, 9

8 6, 2, 8, 2, 6, 5

4 6, 1, 8, 6, 3

9 red, blue, red, green,  
blue, red

5 10, 12, 11, 10, 15

10 20, 25, 22, 25, 21

**Level 2 · Building up****WE DO**

With your teacher: copy what they write on the board.

**Example 3**

Find the mean and mode of: 5, 9, 8, 6, 9, 7. Round the mean to 2 decimal places.

**Example 4**

Find the mean and mode of the scores in this table.

| Score     | 1 | 2 | 3 | 4 |
|-----------|---|---|---|---|
| Frequency | 2 | 5 | 4 | 1 |

**YOU DO**

On your own: Set C. Find the mean and the mode. Round the mean to 2 decimal places where needed.

- 1 5, 3, 2, 6, 7, 5, 5

- 6 Goals in 8 games: 2, 0, 3, 1, 2, 4, 2, 2

- 2 9, 5, 2, 2, 5, 2, 4, 7

- 7 Pocket money (\$): 15, 20, 10, 25, 20, 30

- 3 12, 15, 11, 15, 10, 14

- 8 Scores: 1 three times, 2 four times, 3 twice, 4 once

- 4 2.5, 3.5, 4, 6

- 9 Goals: 0 twice, 1 four times, 2 three times, 3 once

- 5 Temperatures ( $^{\circ}\text{C}$ ) for a week: 21, 24, 19, 24, 22, 23, 21

- 10 Heights (cm): 150, 145, 160, 155, 148, 152



## Level 3 · Challenge

**WE DO**

With your teacher: copy what they write on the board.

**Example 5**

The mean of five numbers is 8. Four of them are 6, 9, 5 and 10. Find the fifth number.

**Example 6**

Ages at a party: 11, 12, 12, 13, 12, 44. Find the mean and mode. Which better describes a typical age?

**YOU DO**

On your own: Set D. Show your working.

- 1 The mean of four numbers is 6. Three of them are 4, 7 and 8. Find the fourth.

- 3 Write a set of five numbers with a mean of 6 and a mode of 4.

- 2 The mean of five test scores is 72. Four are 68, 75, 70 and 80. Find the fifth.

- 4 For 3, 5, 6, 6, 40, find the mean and mode. Which is the better measure? Why?

**YOU DO** On your own: Set D continued. Show your working.

- 5 The number 20 is added to 4, 6, 7, 8, 10. How does the mean change?

- 8 Shoe sizes sold: 7, 8, 8, 9, 10, 8, 11. Should the shop use the mean or the mode to decide what to order? Why?

- 6 Mia scored 7, 8, 6 and 9 on four quizzes. What must she score next to have a mean of 8?

- 9 The mean of three whole numbers in a row (like 4, 5, 6) is 17. What are the numbers?

- 7 The mean of six numbers is 15. One is removed and the mean of the other five is 14. Which number was removed?

- 10 Class A: 12, 15, 15, 18, 20. Class B: 10, 14, 16, 19, 21. Compare their means and modes.

 **Extension · Stretch yourself**

Finished Level 3? Stretch yourself here.

 **Worked example**

The mean age of four friends is 13. A fifth friend joins and the mean age becomes 14. How old is the new friend?

1. Total of four ages:  $4 \times 13 = 52$
2. Total of five ages:  $5 \times 14 = 70$
3. New friend:  $70 - 52 = 18$
4. Check:  $70 \div 5 = 14 \checkmark$

**Key idea:** mean  $\times$  number of values = total.

**Answer:** 18 years old

**YOU DO**

On your own: read the worked example, then try these.

- E1** The mean mark of 10 students is 65. One test is re-marked and gains 15 marks. What is the new mean?

- E2** The mean of 8 numbers is 12. The numbers 5 and 9 are removed. What is the mean of the rest?

- E3** Can the mean be a number that is not in the data? Give an example.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

 **Level 1**

Find the mean of: 4, 6, 8, 10, 12

 **Level 2**

Find the mean and mode of: 3, 7, 7, 2, 9, 5

 **Level 3**

The mean of four numbers is 9. Three of them are 7, 12 and 5. Find the fourth number.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Mean** = total of the values  $\div$  number of values.
  2. **Mode** = the value that appears most often.
  3. There can be **no mode** or **more than one** mode.
  4. An **outlier** pulls the mean towards it; the mode is not affected.

## Level 1

Find the mean and mode of 3, 5, 7, 5, 10

1. Total:  $3 + 5 + 7 + 5 + 10 = 30$
2. 5 values, so  $30 \div 5 = 6$
3. 5 appears most

Answer: mean 6, mode 5

## Level 2

Find the mean and mode of 7, 8, 6, 9, 9, 5, 8

1. Total = 52, 7 values
2.  $52 \div 7 = 7.428... \approx 7.43$
3. 8 and 9 both appear twice

Answer: mean 7.43, modes 8 and 9

## Level 3

The mean of 4 scores is 15. Three are 12, 18, 14. Find the fourth.

1. Total needed:  $4 \times 15 = 60$
2. Known total: 44
3.  $60 - 44 = 16$

Answer: 16

## Extension

5 numbers have a mean of 10. A sixth is added and the mean is 12. What was added?

1.  $5 \times 10 = 50$
2.  $6 \times 12 = 72$
3.  $72 - 50 = 22$

Answer: 22

This space is left blank on purpose. It is the back of your exit ticket.